

Laparoscopic Cholecystectomy Technique

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TECHNIQUE

Approach Considerations

An important issue for surgeons performing a laparoscopic cholecystectomy is whether and when the procedure should be converted to an open cholecystectomy. In the following situations, a low threshold for conversion to an open procedure should be maintained ^[61]:

- Excessive uncontrollable bleeding is encountered
- Patient anatomy is unclear
- Multiple tubular structures are seen entering the “gallbladder,” or a very large cystic duct is seen (especially if it appeared normal on ultrasonography [US]; these findings suggest that the surgeon may be in the wrong place—that is, what is perceived as the cystic duct may actually be the common bile duct [CBD])

Conversion to an open procedure should not be considered a complication, and the possibility that it will prove necessary or advisable should be discussed with the patient preoperatively. In most series, conversion rates are higher with emergency operations. Reported rates range from 1.5% to 15%, with most studies reporting rates around 5% in elective cases. ^[13]

A multivariate analysis identified male sex, elevated white blood cell count, low serum albumin, pericholecystic fluid noted on US, diabetes mellitus, and elevated total bilirubin as independent predictors of conversion. Another multivariate analysis identified male sex, positive Murphy sign, gallbladder-wall thickness exceeding 4 mm, and previous upper abdominal surgery as independent predictors of conversion to an open procedure.

The decision to convert to open cholecystectomy should be made when important anatomic structures cannot be clearly identified or when no progress is being made. As a general rule, if the junction of the gallbladder and the cystic duct has not been identified within 30 minutes of the start of the procedure, a laparoscopic cholecystectomy should be converted to an open cholecystectomy.

Several surgical societies have produced guidelines for safe cholecystectomy. ^[62] The Society of Endoscopic and Laparoscopic Surgeons of India (SELSI) has also published a Consensus Statement for Safe Cholecystectomy. ^[63, 64]

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Conventional Laparoscopic Cholecystectomy

The skin is initially prepared with chlorhexidine from just below the nipple line to the inguinal ligaments and laterally to the anterior superior iliac spines.^[41] The operative field is then draped with sterile drapes.

Placement of ports and instruments

A 1.5-cm longitudinal incision is made at the inferior aspect of the umbilicus, then deepened through the subcutaneous fat to the anterior rectus sheath. A Kocher clamp is used to grasp the reflection of the linea alba onto the umbilicus and elevate it cephalad.

A 1.2-cm longitudinal incision is made in the linea alba with a No. 15 blade. Two U stitches, one on either side of the fascial incision, are placed with 0 polyglactin suture on a curved needle (see the video below).



Laparoscopic cholecystectomy. Placement of fascial stay sutures.

The peritoneum is elevated between two straight clamps and incised so as to afford safe entry into the abdominal cavity. An 11-mm blunt Hasson trocar is placed into the abdominal cavity, and insufflation of carbon dioxide is initiated to a maximum pressure of 15 mm Hg.

The author prefers a 30° laparoscope to a 0° laparoscope on the grounds that it gives better visualization of the cystic structures from multiple vantage points. A 30° scope, however, requires a more skilled scope operator.

The laparoscope is white-balanced and advanced slowly into the abdominal cavity. A 1.2-cm incision is made three fingerbreadths below the xiphoid process and deepened into the subcutaneous fat. An 11-mm trocar is advanced into the abdominal cavity under direct vision (see the image below) in the direction of the gallbladder through the abdominal wall, with care taken to enter just to the right of the falciform ligament.



Laparoscopic cholecystectomy. Advancement of 11-mm trocar under direct vision.

The table is then adjusted to place the patient in a reverse Trendelenburg position with the right side up to allow the small bowel and the transverse colon to fall away from the operative field (see the image below).



Laparoscopic cholecystectomy. Visualization of gallbladder after placement of table in reverse Trendelenburg position.

A 5-mm grasper is placed through the 11-mm subxiphoid port and applied to the fundus of the gallbladder. The gallbladder is then elevated cephalad over the dome of the liver to facilitate the surgeon's choice of the optimal positions for the lateral 5-mm ports.

After appropriate port sites are chosen, the lateral skin incisions are made, and two 5-mm trocars are advanced into the peritoneal cavity under direct vision (see the first image below). A 5-mm grasper with locking mechanism is placed through one of these lateral ports to hold and retract the gallbladder fundus (and thus the liver) (see the second image below).



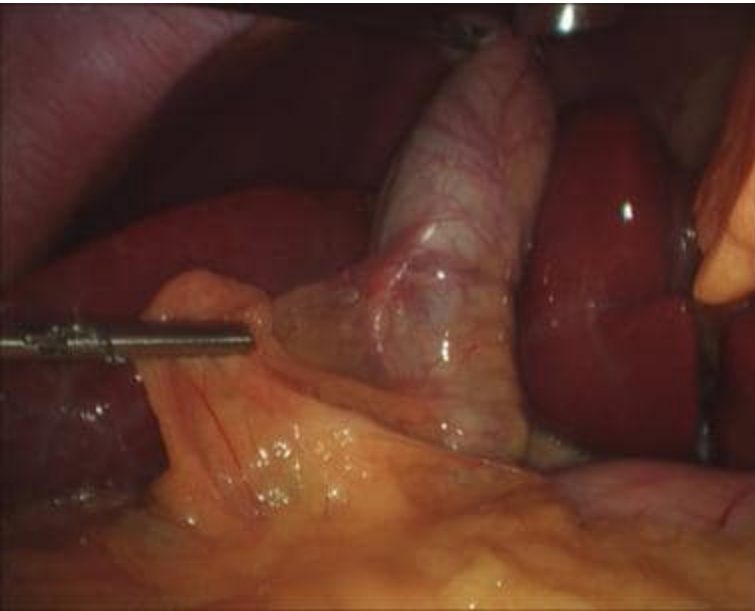
Laparoscopic cholecystectomy. Placement of two lateral 5-mm ports under direct vision.



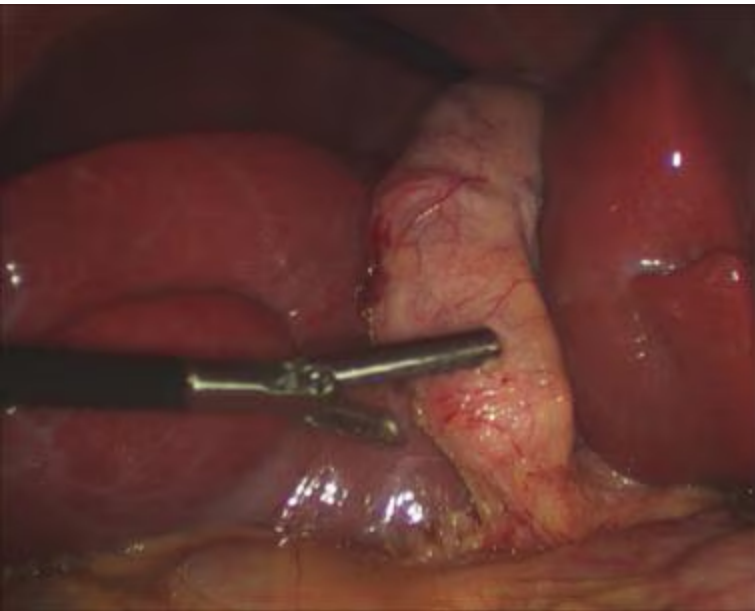
Laparoscopic cholecystectomy. External view after port placement.

Exposure and dissection

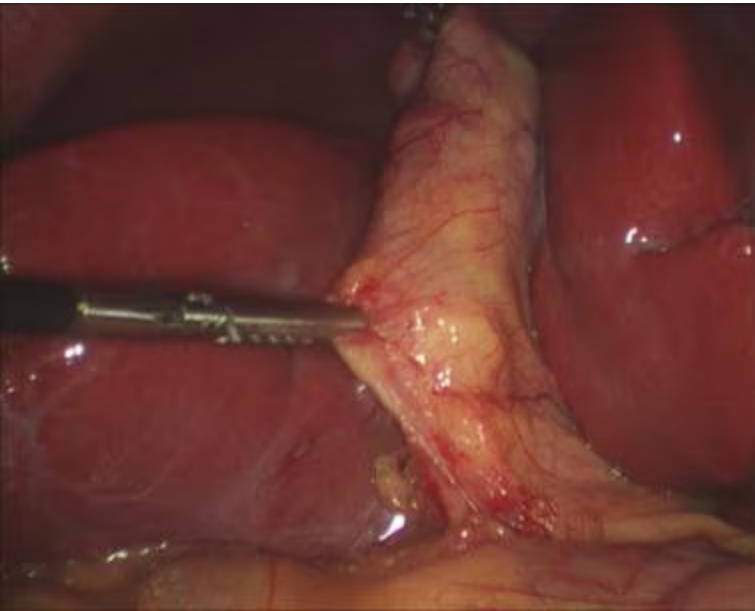
The lateral grasper is applied to the fundus and used to hold it cephalad over the dome of the liver (see the first image below); the medial grasper is used to retract the infundibulum caudolaterally (see the second and third images below). This maneuver straightens the cystic duct (ie, retracts it at 90° from the CBD) and helps protect the CBD from inadvertent injury.^[65] In contrast, cephalad retraction of the infundibulum tends to bring the cystic duct in line with the CBD, making the CBD more prone to injury as a consequence of being misidentified as the cystic duct.



Laparoscopic cholecystectomy. Lateral grasper is used to retract fundus cephalad and retract adhesions.



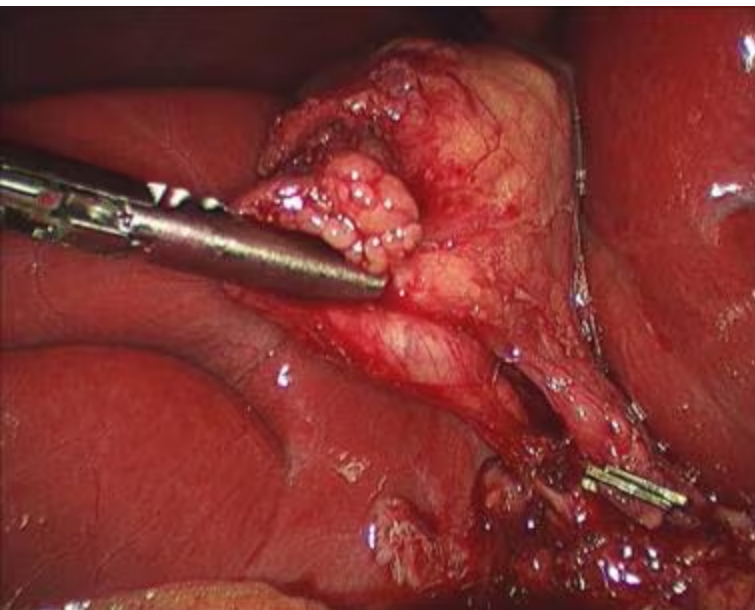
Laparoscopic cholecystectomy. Medial grasper is applied to infundibulum.



Laparoscopic cholecystectomy. Medial grasper is used to retract infundibulum in caudolateral direction.

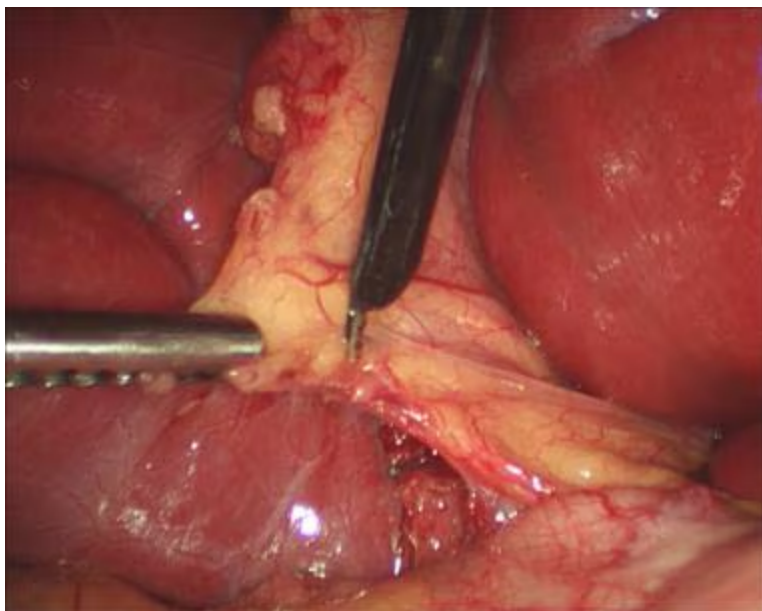
Occasionally, adhesions are encountered between the gallbladder and the omentum or duodenum. These may be carefully lysed with a hook cautery. The authors prefer to use an L-hook electrocautery, which allows a very clean and delicate dissection, but any electrosurgical device can be used for this purpose.

Once the area of the hilum of the gallbladder has been reached, the importance of exposure and delicate dissection cannot be overemphasized. The cystic duct and artery must be carefully dissected and identified in the triangle of Calot to obtain the "critical view of safety" (CVS). This critical view is achieved when the surgeon can see only two structures (the cystic duct and artery) entering directly into the gallbladder (see the image below); it must be obtained before any structures are clipped or transected. [66]



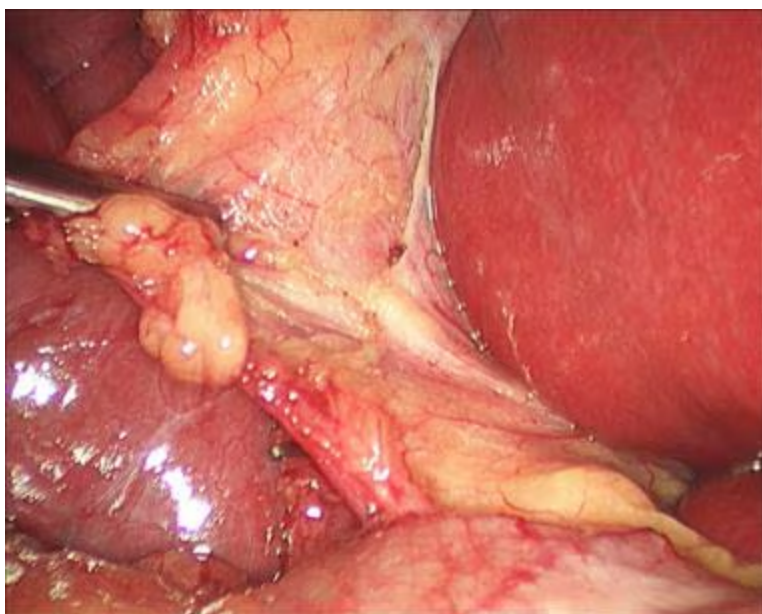
Laparoscopic cholecystectomy. Critical view, with only cystic duct and cystic artery seen entering gallbladder.

The key to obtaining the CVS is complete clearance of the areolar tissue in the subhepatic space. With the infundibulum held caudolaterally, the hook is used to score the anterior (medial) peritoneum overlying the infundibulum–cystic duct junction (see the image below).



Laparoscopic cholecystectomy. Use of L-hook electrocautery to score anterior peritoneum.

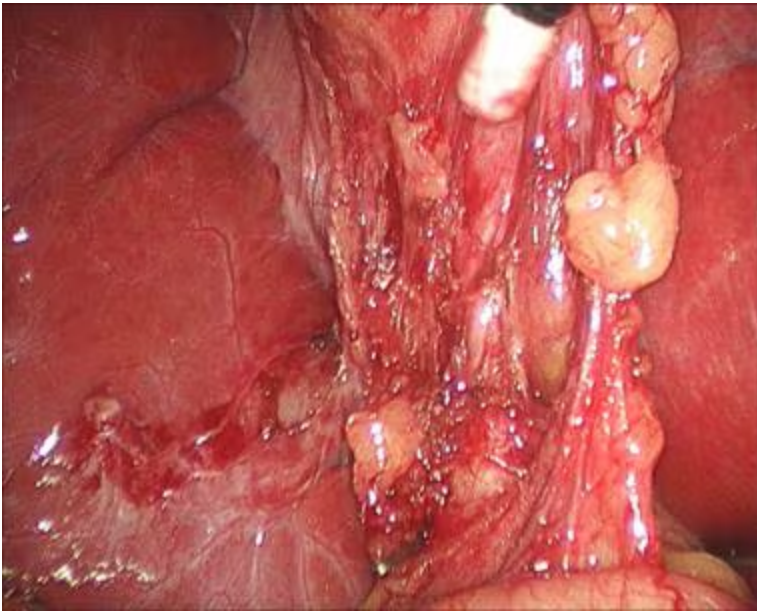
Next, the peritoneum is incised along the medial aspect all the way to within 1 cm of the liver (see the image below), and the incision is continued cephalad toward the fundus of the gallbladder.



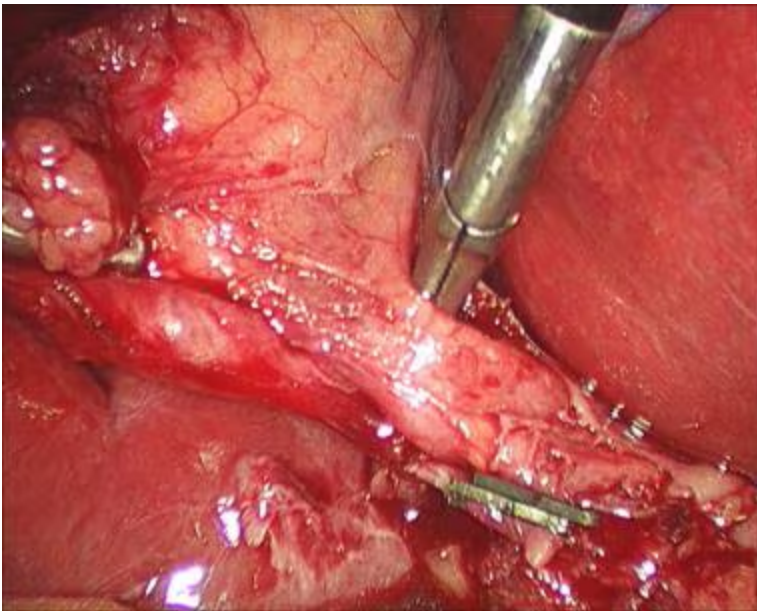
Laparoscopic cholecystectomy. Division of peritoneum along medial aspect.

The gallbladder is then retracted caudomedially, and a similar dissection is carried out on the lateral (posterior) surface. This technique is sometimes referred to as the flag technique. ^[1]

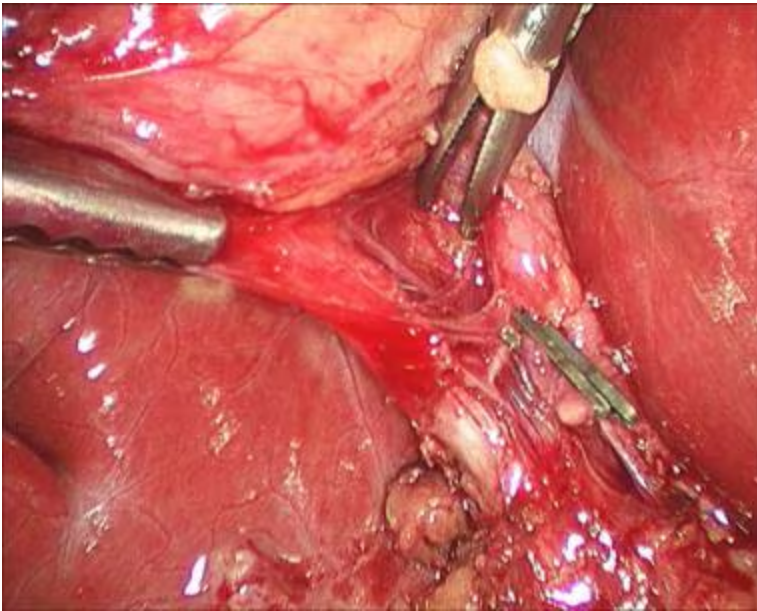
An Endo Peanut or a Maryland dissector can be of great help in better defining these structures (see the images below). At this point, the surgeon should be able to identify the cystic duct and cystic artery entering directly into the gallbladder; this is the CVS.



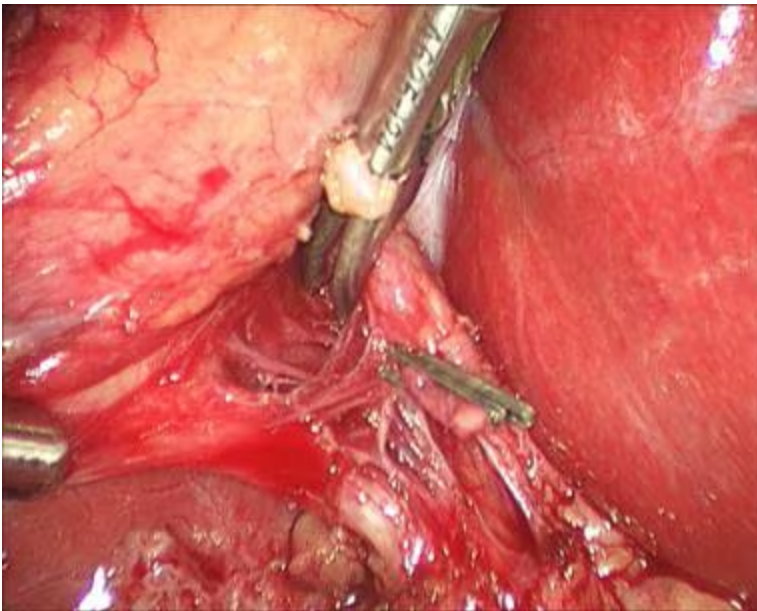
Laparoscopic cholecystectomy. Use of Endo Peanut to identify cystic structures.



Laparoscopic cholecystectomy. Use of Maryland dissector to dissect cystic duct.



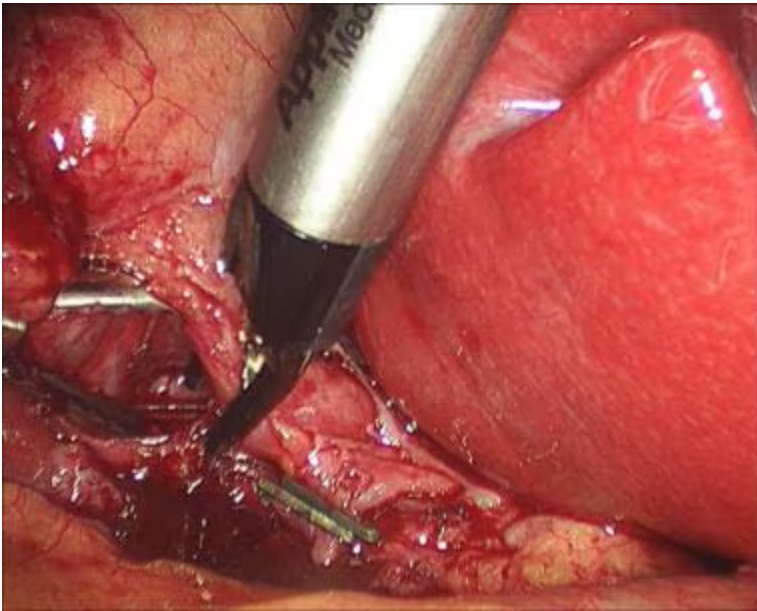
Laparoscopic cholecystectomy. Use of Maryland dissector to dissect cystic artery.



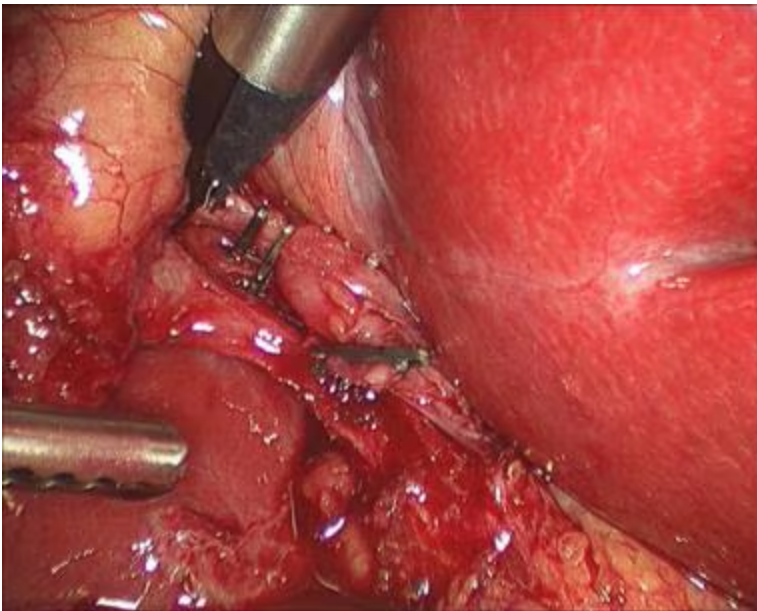
Laparoscopic cholecystectomy. Continued dissection of critical structures.

Clipping and division of cystic structures

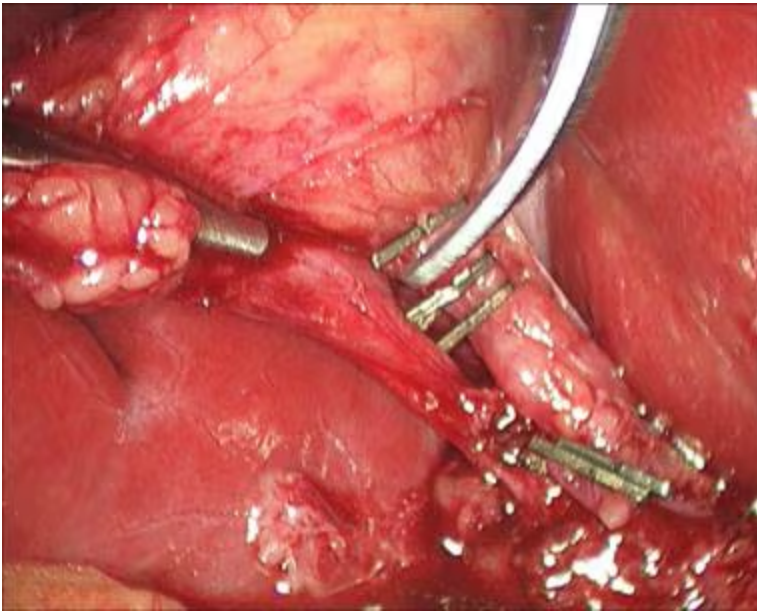
Once the CVS has been achieved and the cystic structures clearly identified, the structures can be clipped and divided (see the images below). An endoscopic clip applier is used to place clips on the artery and duct (two proximally and one distally), which are then divided with endoscopic shears (eg, Endo Shears; Covidien, Mansfield, MA).



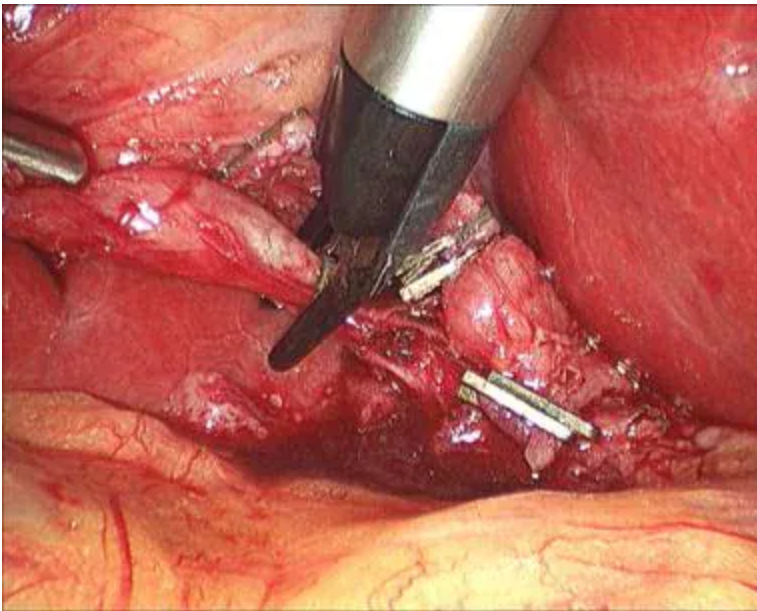
Laparoscopic cholecystectomy. Placement of clip at lower aspect of cystic artery.



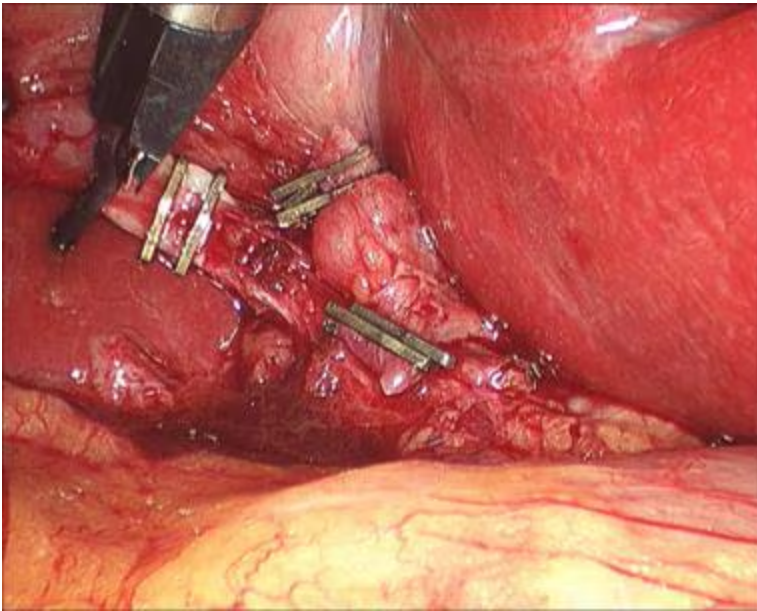
Laparoscopic cholecystectomy. Placement of superior clips on cystic artery.



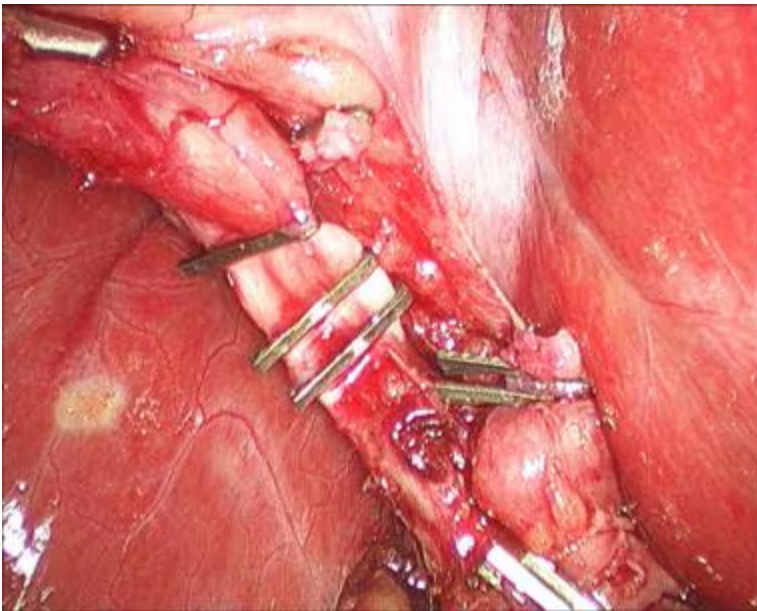
Laparoscopic cholecystectomy. Transection of cystic artery with Endo Shears.



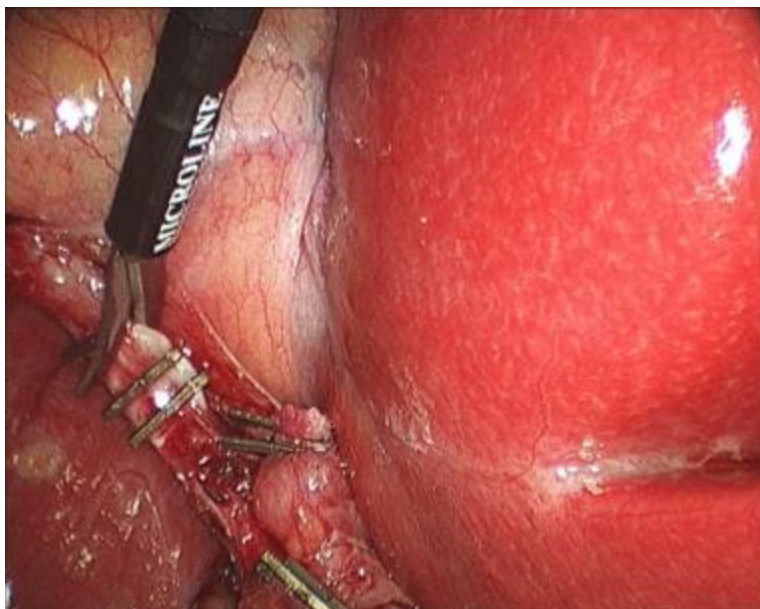
Laparoscopic cholecystectomy. Placement of clips on distal cystic duct.



Laparoscopic cholecystectomy. Placement of proximal clip on cystic duct.



Laparoscopic cholecystectomy. View of clipped cystic duct before transection.

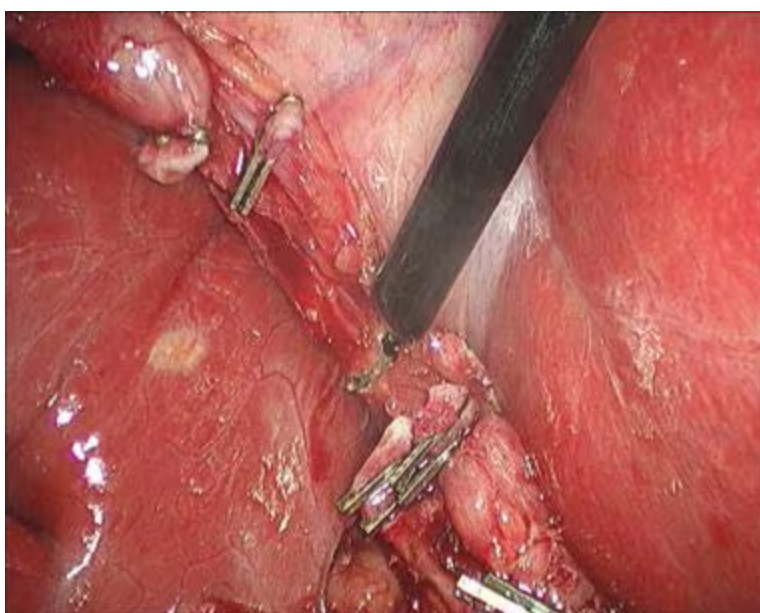


Laparoscopic cholecystectomy. Transection of cystic duct between clips with Endo Shears.

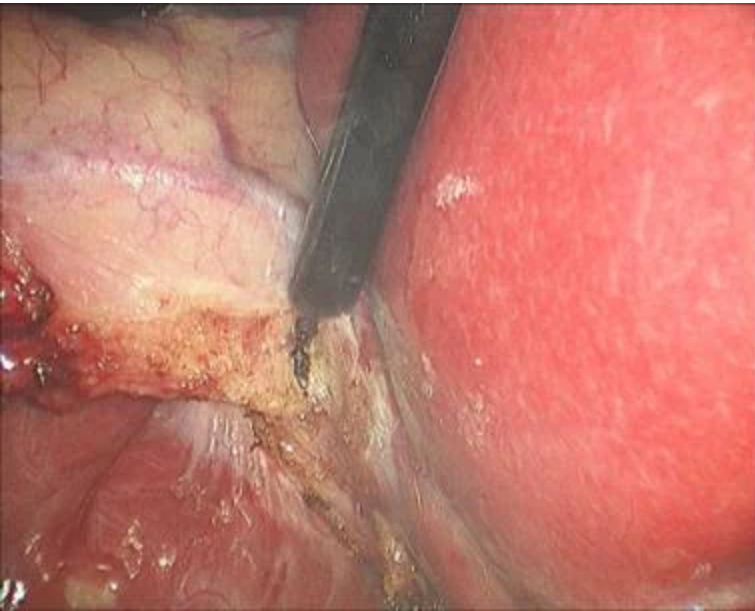
When the cystic duct is large, there are several options that may be considered, including an endoscopic stapler, Endoloops, and the overlapping clip technique. [67, 68, 69] If an Endoloop is used, it should be placed through the subxiphoid port. The grasper from the infundibulum is used to gently hold the cystic stump through the loop; the loop is then pulled taut.

Mobilization and removal of gallbladder

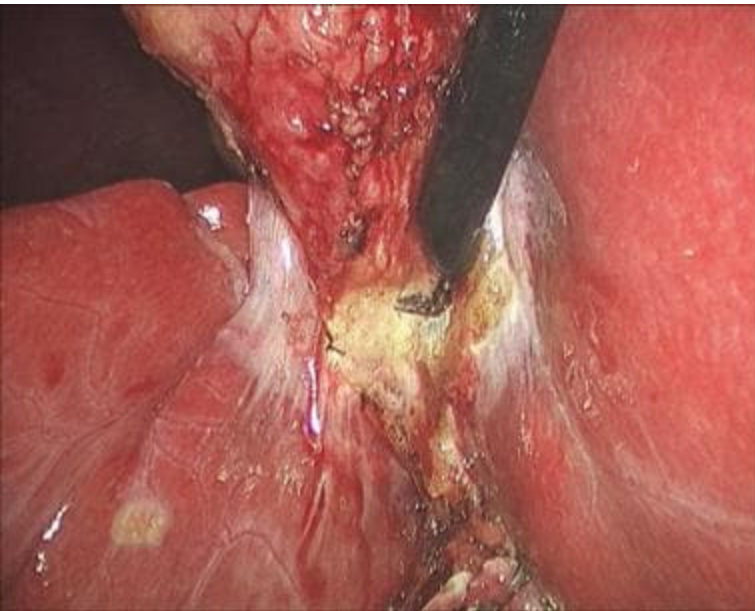
Once the cystic structures have been clipped and divided, the infundibulum is retracted cephalad. A hook or spatula is used to develop a plane in the areolar tissue between gallbladder and liver (cystic plate) with smooth sweeping movements from right to left and back again. As in any surgical procedure, the traction-countertraction rule is essential. As the dissection marches up the gallbladder bed, the assistant should reposition his or her graspers to ensure optimal tension on the areolar tissue between gallbladder and liver bed (see the images below).



Laparoscopic cholecystectomy. Use of hook to develop plane in areolar tissue between gallbladder and liver.



Laparoscopic cholecystectomy. Use of traction and hook to remove gallbladder from gallbladder bed.



Laparoscopic cholecystectomy. Side-to-side sweeping motion with electrocautery to remove gallbladder from gallbladder bed.

It is important to be alert for any aberrant vessels and ducts that may arise from the liver bed and enter directly into the gallbladder. These should be clipped and not simply cauterized.

Before the last strands connecting the gallbladder to the liver are divided, a final inspection of the gallbladder fossa and the clipped cystic structures should be carried out. Any bleeding points in the gallbladder fossa should be controlled at this time, before the gallbladder is completely separated from the liver (see the image below). This is the surgeon's last opportunity to visualize these areas well.



Laparoscopic cholecystectomy. Cauterization of any bleeding in gallbladder bed before complete division of gallbladder.

Both 5-mm graspers are applied to the gallbladder and used to hold it over the right upper quadrant. An endoscopic retrieval pouch is inserted into the peritoneal cavity through the subxiphoid trocar. The gallbladder is placed in the bag, which is then cinched closed and extracted through the subxiphoid port. (See the images below.) Some surgeons transfer the laparoscope to the subxiphoid port and place and extract the bag through the umbilical port. A final inspection and washout is performed.

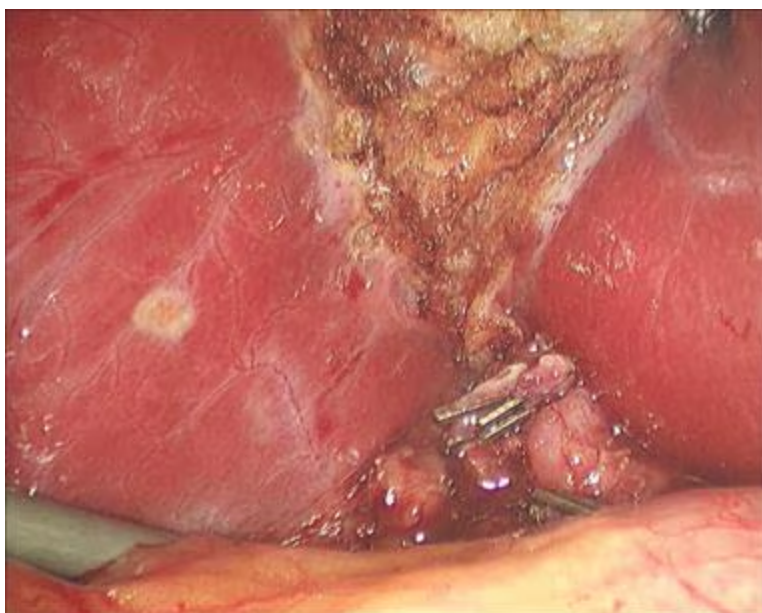


Laparoscopic cholecystectomy. Placement of gallbladder into endoscopic retrieval pouch.



Laparoscopic cholecystectomy. Placement of gallbladder into endoscopic retrieval pouch and removal of instrument from pouch.

The table is returned to the neutral position. The gallbladder bed and the suprahepatic spaces are irrigated and suctioned to ensure adequate hemostasis and removal of any debris or bile that may have spilled (see the image below).



Laparoscopic cholecystectomy. Irrigation and suction of gallbladder bed.

Port removal and closure

The subxiphoid port and the two 5-mm ports are removed under direct vision, followed by the Hasson trocar (see the image below).



Laparoscopic cholecystectomy. Removal of ports under direct vision.

Overall, the incidence of port-site hernias is very low. Tonouchi et al reported an incidence of 0.65-2.8%; their recommendation was to close all port sites larger than 10 mm. ^[70] Some surgeons, however, do not close the subxiphoid port. The fascia is closed at the umbilical port by using the two U stitches placed at the beginning of the procedure. All of the skin incisions are closed with 4-0 absorbable monofilament suture, followed by cyanoacrylate tissue adhesive. (See the image below.)



Laparoscopic cholecystectomy. Abdomen after skin closure.

Postoperative care

Finally, the patient is extubated, transferred to the postanesthesia care unit, and monitored for 4-6 hours. In cases of elective cholecystectomy, the patient can be discharged home with a combined oral pain medication containing acetaminophen and an opiate. ^[3]

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Alternative Minimally Invasive Approaches

Minilaparoscopic cholecystectomy

Since the introduction and widespread adoption of laparoscopic cholecystectomy, investigators have attempted to make further improvements to the established technique, aiming to reduce the invasiveness of the procedure by decreasing the number and, more commonly, the size of the operating ports, as well as the size of the instruments.

Novitsky et al concluded that laparoscopic cholecystectomy can be safely performed with the use of 10-mm umbilical, 5-mm epigastric, 2-mm subcostal, and 2-mm lateral ports. They suggest that minilaparoscopy can be routinely offered to properly selected patients undergoing elective laparoscopic cholecystectomy. [71]

Single-site cholecystectomy

Cholecystectomy can also be performed through single-port access (SPA) surgery, also known as single-incision laparoscopic surgery (SILS)—so-called single-port cholecystectomy. [72, 73] In these procedures, the surgeon operates almost exclusively through a single entry point, typically the patient's umbilicus.

In a retrospective review that included 360 consecutive patients who underwent single-incision laparoscopic cholecystectomy (SILC), Sato et al concluded that SILC appeared to be relatively safe and to have an acceptable postoperative complication rate but noted that poor physical status and acute cholecystitis were significant risk factors for postoperative complications. [74]

An approach in which three discrete ports are placed through three mini-incisions at a single site has also been described. [75]

Robotic cholecystectomy

Robotic-assisted cholecystectomy is feasible and safe; however, its cost is high, and no clear benefit for patients has been established. At present, therefore, the use of this technology is not justified. [76, 77]

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Complications

Hasson trocar/Veress needle injury

Intestinal injury may occur during establishment of abdominal access, adhesiolysis, or dissection of the gallbladder away from the duodenum or colon. An injury to the bowel should be repaired with careful one- or two-layer suture closure.

The rates of injury to viscera or vessels from a Hasson trocar or from a Veress needle are comparable (in the range of 0.2%).

Hemorrhage

When a large-vessel vascular injury occurs, it is usually at the time of initial abdominal access. Such injuries may be lethal complications. Development of a retroperitoneal hematoma or hypotension should be treated immediately by converting to laparotomy.

Excessive bleeding in the region of the triangle of Calot should not be treated laparoscopically. In this situation, attempts at blind clipping or cauterization usually lead to worsening hemorrhage or hepatic artery injury. If, and only if, a bleeding site can be definitely identified and the locations of both the hepatic artery and the CBD are known, bleeding may be controlled with cauterization or clipping.

Bleeding in the gallbladder bed can usually be controlled by fulgurating the bleeding site. The author prefers using a spatulated electrocautery wand for this purpose. If a larger intrahepatic sinus has been entered, hemostatic agents (eg, microfibrillar collagen) can be placed laparoscopically in the gallbladder bed, and pressure can be maintained with a clamp. The argon plasma coagulator (APC) can be an excellent tool for severe gallbladder fossa oozing that is not responsive to simple electrocauterization.

Postcholecystectomy syndrome

The term postcholecystectomy syndrome refers to a set of abdominal symptoms that occur in as many as 40% of patients after cholecystectomy. Symptoms are often vague and include dyspepsia, flatulence, bloating, right-upper-quadrant pain, and epigastric pain. The most common causes of this syndrome are dietary indiscretion, retained CBD stones, inflammation of the cystic duct remnant, and sphincter of Oddi dysfunction. [78]

Bile duct injury

The most dreaded complication of laparoscopic cholecystectomy is bile duct injury (BDI; ie, injury to the CBD or the common hepatic duct [CHD]). The estimated incidence of BDI in cholecystectomies performed laparoscopically ranges from 0.3% to 2.7%. By way of comparison, BDIs have been estimated to occur in 0.25-0.5% of open cholecystectomies.

A major risk factor for BDI is relative inexperience on the part of the surgeon. [79, 80, 81] Other risk factors are the presence of aberrant biliary tree anatomy and the presence of local acute or chronic inflammation.

BDI may manifest as bile leakage (which is due to partial or complete bile duct transection leading to passage of bile into the peritoneum) or as biliary obstruction (which may be either partial or complete and is secondary to acute ductal ligation or chronic stricture formation).

BDI may present at different points in the treatment of the patient, as follows:

- Intraoperative presentation - The injury is identified during the index procedure (ie, laparoscopic cholecystectomy)
- Delayed presentation - Patients may present 3-7 days after surgery with fever, abdominal pain, anorexia, ileus, ascites, nausea, or jaundice
- Late-onset stricture - Patients may present months later with abdominal pain and jaundice

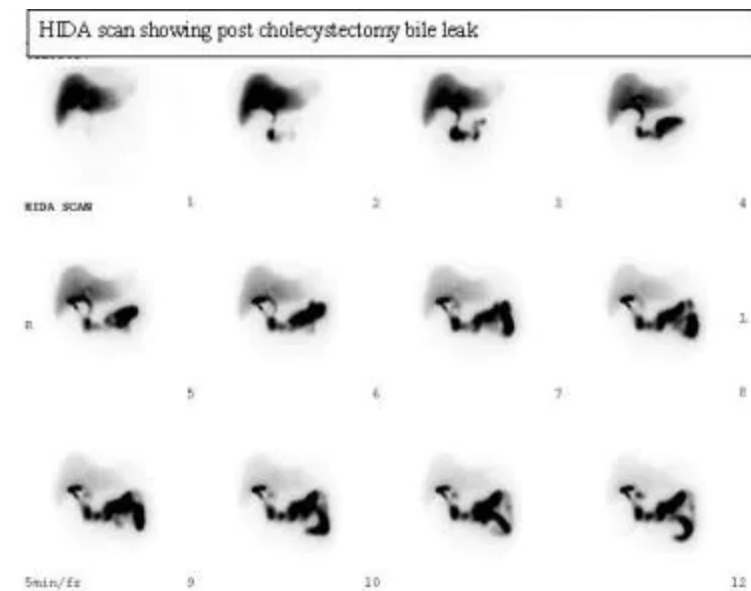
Management of BDI depends on the timing of identification and the degree of injury and the expertise and experience of the surgeon in reconstructive biliary surgery. If the injury is identified intraoperatively, the injured duct should be repaired immediately. Depending on degree, most partial CBD injuries can be primarily repaired over a T-tube, [82] though in some cases, repair may require transfer to a specialized center.

If the injury is identified postoperatively, imaging (US, computed tomography [CT], or both) is indicated to look for collections or ductal dilatation (see the first image below). Biliary scintigraphy with ^{99m}Tc-IDA or hepatobiliary iminodiacetic acid (HIDA) can be used to diagnose and sometimes localize the source of bile leakage (see the second image below). Endoscopic retrograde

cholangiopancreatography (ERCP) can be used both diagnostically and therapeutically (see the third and fourth images below).



Laparoscopic cholecystectomy. CT scan illustrating biloma.



Laparoscopic cholecystectomy. HIDA scan showing postcholecystectomy leak.

ERCP showing post cholecystectomy leak of contrast



Laparoscopic cholecystectomy. Postcholecystectomy ERCP showing leak of contrast.

Post ERCP image shows stent in place



Laparoscopic cholecystectomy. Postcholecystectomy ERCP.

Treatment of BDI can be an involved subject but may be summarized as follows:

- CT-guided drainage of biloma followed by ERCP with endoscopic sphincterotomy and stent placement (see the image below) is the treatment of choice for less severe lesions, such as minor lacerations of the CBD, duct of Luschka leak, or displaced cystic duct clips
- Lesions in the proximal biliary tree may be more amenable to percutaneous transhepatic approaches
- Surgical biliary reconstruction may be necessary in cases of complete BDI



Laparoscopic cholecystectomy. ERCP-guided stent placement.

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